

Introduction: What are quasi-RCTs – how are they different from regular RCTs?

A quasi-experiment is a research design that, while resembling an experiment, **does not fulfill all the criteria of a true experiment**. They share similarities with controlled experiments, such as comparing different conditions (independent variable), yet they typically lack the **random assignment of subjects to different conditions**. This also means that a researcher typically has not designed and carried out the manipulation. Keep in mind, that in a regular RCT, a researcher would have randomly assigned participants to different groups, and the researcher would have designed and carried out the manipulation (who gets what incentive, for example). Such a random assignment and manipulation of treatment ensures that any differences observed between the groups are due to the treatment intervention rather than pre-existing characteristics of participants or other confounding variables. Instead of relying on randomization, quasi-experiments take advantage of natural or existing conditions where people have ended up in different conditions, but without control over how participants end up in different groups.

In other words, while an RCT would randomly assign participants to receive or not to receive a treatment, a quasi-experiment compares groups that were naturally exposed to different treatments, while assuming the groups are similar enough to draw meaningful comparisons. This reliance on **assumptions** introduces the possibility of **confounding variables**—factors that might influence the outcome independently of the treatment. This explanation is somewhat abstract, which is why I present a few examples further below, to illustrate how such quasi-experiments might look like in practice.

Quasi-experiments are especially valuable in **real-world settings**, such as in management or economics, where randomization may be impractical or unethical. For example, if one is interested in how to manage an organizational change program, one is very unlikely to be able to randomize who gets access to which program across 100s of organizations in actual need of organizational change. Quasi-experiments provide a practical way to study the effects of interventions in complex, uncontrolled environments, but the absence of randomization means they can only approximate causality, rather than definitively establish it as RCTs can.

Example 1

Imagine a government job-training program that accepts people with a score above 60 (scale from 1 to 100, normally distributed). In a study, researchers look at two groups that have very similar results, and thus seem to have similar abilities: people who scored just over 60 (e.g. 60.2) and those who scored just under 60 (e.g. 59.8). Due to noise in the assessment of people, the difference between 60.2 and 59.8 is at most very small and possibly just random. Measurement of performance is not always exact, and small differences might therefore not reveal actual differences. In an actual experiment, one would have randomized access to job training. Here we have to assume that such small differences in scores are in effect meaningless, and that the groups are - almost - identical. By comparing the following job performance of these groups (those just above and below 60), the study aims to see if the training really helps in getting better jobs or higher salaries. This method is useful because it compares people who are very similar, except for being in or out of the program, helping to understand the program's true effect. If those that get a rating of 60.2 (and thus get job training) in the future perform much better than those at 59.8 (without job training), one has evidence

supporting the value of the job training. Remember, the intervention (job-training) is not randomly allocated, and even the small differences in rating might be meaningful, and thus be a confounding variable. Quasi-experiments thus rely on assumptions – here the assumption is that the difference is not relevant.

Example 2

For another example, consider an actual study on cognitive fatigue carried out on Danish elementary school students, where researchers analyzed test performance data from several years at all Danish schools (example from <https://doi.org/10.1073/pnas.1516947113>). Such elementary school kids take tests at different times of the day (e.g. early morning or late afternoon), and researchers compared the results of kids who took tests at different times of the day. Results show that taking the tests earlier improved performance. Importantly, the timing of these tests was not actually randomly assigned (as in a system administrator flipping a coin). Rather, the allocation of when to do a test was simply based on when what rooms and classes were available. Hence, researchers could argue and assume that the allocation of time slots was as good as random. Now, if teachers had systematically allocated good students to morning time slots, then the result (morning time slots lead to better results) would have been a confounding factor and the actual driver of differences in performance levels at different test times. The argument thus is that there was no systematic relation between expectations of ability and when a kid did a test. Exploiting the natural allocation of time slots at schools in this way means that researchers do not have to set up a large, complicated, resource-intensive collaboration with a range of schools to randomly assign students to certain test times. Overall, such a quasi-experimental approach provided a comprehensive, realistic view of how timing of tests affects performance in a real-world educational setting.

Example 3

In contrast, consider a scenario where managers at an organization roll out a new bonus system and want to assess its impact on employee productivity. This system was deliberately only rolled out to half of the employees (not randomly selected) in the beginning, seemingly creating a natural setup for a quasi-experiment. However, imagine this rollout happened at the same time as other significant changes, such as an office relocation or job-training affecting some of the employees. These other changes might be confounding variables, meaning any improvement in performance might be due to these other factors, rather than the treatment (bonus system). For example, maybe more of the employees in the treatment group (compared to control) received additional job-training, and the employees might have improved in performance due to the job-training, rather than the bonus system. With multiple changes in the workplace being potential causes, it is very difficult to statistically isolate causal effects. Consequently, despite its first appearance as a quasi-experiment, the presence of these confounding variables prevents a clear understanding of the bonus system's direct impact.

To sum up

While quasi-experiments cannot establish causality as definitively as randomized controlled trials, they are still very useful in investigating relationships in complex, real-world contexts. Their flexibility and practicality make them a popular choice in many fields, especially when ethical or logistical constraints limit the possibility of true experiments.